Cybersecurity - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The AH Protocol provides source authentication and data integrity, but not
   - (a) Integrity
   - (b) Privacy
   - (c) Nonrepudiation
   - (d) Both A & C
2. The digest created by a hash function is normally called a
   - (a) Modication detection code (MDC)
   - (b) Modify authentication connection
   - (c) Message authentication control
   - (d) Message authentication cipher
3. Which of the following deals with network intrusion detection and real-time traffic analysis?
   - (a) John the Ripper
   - (b) L 0 phtCrack
   - (c) Snort
   - (d) Nessus
4. What was the first buffer overflow attack?
   - (a) Love Bug
   - (b) SQL Slammer
   - (c) Morris Worm
   - (d) Code Red
5. ______________________________ is alike as that of Access Point (AP) from 802.11, & the mobile operators uses it for offering signal coverage.
   - (a) Base Signal Station
   - (b) Base Transmitter Station
   - (c) Base Transceiver Station
   - (d) Transceiver Station

True / False (10 marks)

Answer True or False

6. True or False: The 4-way handshake is a four-step authentication process used when a wireless user connects to an access point.

7. True or False: Snort is primarily designed for network analysis in multiprotocol diverse networks, similar to EtterPeak.

8. True or False: Eavesdropping is a legitimate method of network monitoring and does not qualify as a security exploit.

9. True or False: The Tor browser is essential for accessing the Tor network and utilizing its anonymous services.

10. True or False: A SYN stealth scan does not complete the TCP handshake, making it a less detectable method for port scanning.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the semantic security of the encryption scheme defined by E(k,m) = [r $\leftarrow$ R, output (r, F(k,r) m)] under chosen plaintext attacks, considering the size of R and the implications of key reuse.

12. Discuss the implications of weak or non-existent authentication mechanisms on presentation layer security in cybersecurity, providing examples and analysis of potential vulnerabilities.

End of Paper